

Wavelet Kolmogorov–Arnold Networks for Inter-Patient ECG Heartbeat Classification: A Controlled Multi-Seed Evaluation

Venkateramanan Manivannan^a, Ramanathan Lakshmanan^{b,*}

^a*School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology, Vellore, 632014, Tamil Nadu, India*

^b*Department of IoT, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, 632014, Tamil Nadu, India*

Abstract

Kolmogorov–Arnold networks (KANs) with wavelet edge functions are promoted as inspectable models. They have not been evaluated for ECG heartbeat classification under the inter-patient protocol with repeated runs and matched baselines. We evaluated a 153,045-parameter wavelet KAN (PC-WavKAN) on MIT-BIH under the de Chazal DS1/DS2 partition against four baselines trained under the same protocol and budget. The design used 20 paired seeds, architecture selection on validation data only, Holm–Bonferroni correction, effect sizes and confidence intervals. DS2 Macro-F1 was 0.357 ± 0.019 and not statistically distinguishable from any baseline (all Holm-adjusted $p \geq 0.36$; every paired 95% interval within ± 0.03). Supraventricular recall was 0.198 at precision 0.175, and the supraventricular operating point was poor for all five models. On validation, a prior-centred initialisation of the wavelet parameters helped relative to an isotropic one, the mother wavelet did not measurably matter, and self-attention over RR-interval features was harmful. The trained parameters did not support a physiological interpretation: the channel groups are exchangeable, and training moved randomly initialised parameters away from the prior values. Class-balanced sampling raised held-out supraventricular recall from 0.137 to 0.222, but not on the validation partition. Re-evaluating all 100 checkpoints without retraining showed two things. An annotation-derived artefact in the RR features of 9.2% of test beats did not inflate performance. Under preprocessing matched to training, the wavelet KAN transferred worse than both convolutional baselines to the INCART and SVDB databases.

Keywords: Kolmogorov–Arnold networks, wavelets, ECG heartbeat classification, inter-patient evaluation, class imbalance, reproducibility

1. Introduction

The electrocardiogram (ECG) is the reference modality for arrhythmia detection, and manual review of long-term recordings does not scale. Convolutional and attention-based networks now report high beat-classification accuracy [1, 2, 3]. Much of that literature, however, evaluates under an intra-patient protocol in which beats from one recording appear in both training and test sets, which lets a classifier memorise a patient’s morphology and inflates reported accuracy [4, 5, 6]. Under the inter-patient DS1/DS2 protocol of

*Corresponding author.

Email addresses: venkate.ramanan2024@vitstudent.ac.in (Venkateramanan Manivannan), lramanathan@vit.ac.in (Ramanathan Lakshmanan)

de Chazal *et al.* [7], performance drops sharply and minority-class recall in particular collapses.

Kolmogorov–Arnold Networks (KANs) [8] place learnable univariate functions on network edges. Wav-KAN [9] uses wavelets parameterised by a translation and a dilation as those functions and argues that this makes the learned function space more inspectable. KAN-based ECG classifiers have been reported [10], as have learnable-wavelet convolutional networks for arrhythmia classification [11]. We found no wavelet-KAN evaluated under the de Chazal protocol with repeated seeds, protocol-matched baselines and multiplicity-corrected statistics. We also found no test of whether its parameters admit the physiological reading that the wavelet parameterisation invites.

This paper reports such an evaluation. It is framed as a controlled study, not as a method proposal. We ask four questions:

- RQ1** Under the inter-patient protocol and a matched training protocol, does a wavelet KAN differ from standard architectures in five-class Macro-F1?
- RQ2** Which design choices measurably matter, as judged on validation data alone: the initialisation of the wavelet parameters, the choice of mother wavelet, the rhythm encoder, an auxiliary morphology branch, and class-balanced sampling?
- RQ3** Do the trained wavelet parameters support a physiological interpretation?
- RQ4** How sensitive are the conclusions to protocol artefacts and to a change of database?

The evaluated model, PC-WavKAN (153,045 parameters), is a wavelet-KAN morphology branch combined with an interval-feature (rhythm) branch. Its one non-standard ingredient is a prior-centred initialisation of the wavelet translations and dilations (PCWI). Our contributions are empirical:

1. A 20-seed, seed-paired inter-patient comparison against four baselines trained under the same protocol and budget, with architecture selection on validation data only, Holm–Bonferroni correction, effect sizes and confidence intervals. It finds no statistically detectable Macro-F1 difference. Every paired 95% confidence interval includes zero and lies within ± 0.03 , and the S-class operating point is poor for all five models (§4.2, §4.3).
2. A single-variable component analysis on validation data (§4.1). Initialising the wavelet parameters around fixed prior values (PCWI) improves validation Macro-F1 relative to an isotropic initialisation in the configuration in which the components were ablated. The mother wavelet is not measurably consequential. Self-attention over the five interval features is harmful and was removed.
3. A test of the interpretability premise with explicit null models (§4.7). The channel blocks that receive the three priors are exchangeable, so no block can be assigned a waveform component. Training preserves much of the prior-centred structure, but it moves randomly initialised parameters *away* from the prior values, so the parameters do not support a physiological reading.
4. Inference-only sensitivity analyses of all 100 checkpoints.
 - An annotation-derived artefact in the interval features does not inflate performance (§4.6).
 - Under a preprocessing pipeline verified identical to training, the wavelet KAN transfers worse than both convolutional baselines to two external databases (§4.8). An unmatched evaluation of the same checkpoints had suggested the opposite.

A further finding concerns class-balanced sampling. Compared with the inverse-frequency reweighting used by the baselines, it raised DS2 supraventricular recall substantially in two configurations, but this gain did not reproduce on the validation partition for the adopted configuration (§4.4).

2. Related Work

Inter-patient evaluation. Systematic reviews document how prevalent intra-patient evaluation remains and how strongly it inflates reported performance [5, 6]. Bahrami *et al.* quantify that optimism directly [4]. The DS1/DS2 partition of de Chazal *et al.* [7] is the established remedy for MIT-BIH, and we adopt it unmodified. Architectures evaluated under it include CNN–RNN hybrids [2], multiscale attention CNNs [12] and sequence-to-sequence models [13]. Differences in lead selection, beat extraction, AAMI mapping and Q-class handling make published accuracies hard to compare directly, so we retrain our baselines under one protocol (§3.5). We give the most direct external comparison, with the protocol’s own source, in §4.3.

KANs and learnable wavelet parameterisations. KANs [8] use B-spline edge functions. Wav-KAN [9] substitutes wavelets, including the Mexican Hat, with learnable translation and dilation. The wavelet-KAN layer, the Mexican Hat basis and the argument that a wavelet basis is more inspectable are due to that work and are not claimed here. Learning only the scale and translation of wavelet kernels has also been used in convolutional front-ends, for example WaveletKernelNet [14] for machinery fault diagnosis and WavelNet [11], a SincNet-inspired wavelet CNN for arrhythmia classification on a subject-oriented MIT-BIH benchmark. Unlike those convolutional kernels, a Wav-KAN edge function acts on the *amplitude* of an individual input sample, not on a time window, a distinction that matters for interpretation (§3.3). MAK-Net [10] combines spline KAN layers with multiscale convolution and a BiGRU for imbalanced ECG classification. It evaluates under an intra-patient protocol and reports Grad-CAM attributions. Initialisation of KANs is an active question: Rigas *et al.* [15] compare variance-preserving and power-law schemes for spline-KAN coefficients. PCWI addresses a different quantity, the wavelet translation and dilation, and leaves coefficient initialisation at a standard default.

Class imbalance. In a systematic study on image benchmarks, Buda *et al.* [16] found oversampling to be the dominant remedy for class imbalance in convolutional networks. They compared it with undersampling, two-phase training and threshold adjustment. Kang *et al.* [17] compare instance-balanced, class-balanced and progressively balanced sampling for long-tailed recognition. Curriculum approaches vary the sampling distribution or loss weighting over training [18, 19, 20], and adaptive logit adjustment is a logit-level alternative studied for record-level ECG diagnosis [21]. Our training script implements a two-phase schedule of this kind. The selected checkpoint always came from its first, class-balanced phase (§3.4), so what our comparison measures is class-balanced sampling against inverse-frequency reweighting.

Efficient inference. Small memory footprints are argued for on-device ECG inference [22, 23]. We report measured size and x86 CPU latency, and say which hardware class was measured (§4.9).

3. Data and Methods

3.1. Data and Preprocessing

We use the 44 non-paced records of the MIT-BIH Arrhythmia Database [24, 25]. The paced records 102, 104, 107 and 217 are excluded under the protocol. Channel 0 is used, which is the modified lead II (MLII) in every record except 114, where channel 0 is V5. Each record is cleaned over its full length with the `neurokit` method of NeuroKit2, a zero-phase 0.5 Hz fifth-order Butterworth high-pass filter followed by a zero-phase 7-sample moving average, which the library designs for 50 Hz mains. A 360-sample window (1 s at 360 Hz) is then extracted at each annotated beat. The window spans 90 samples (250 ms) before and 270 samples (750 ms) after the annotated R-peak, so the R-peak sits at sample 90. Each window is z -scored with its own mean and standard deviation, and windows with degenerate variance are discarded. No normalisation statistic is estimated across beats, records or splits.

Interval features. Each beat i also receives a five-element vector

$$r_i = [RR_{-2}, RR_{-1}, RR_0, RR_{+1}, RR_{+2}], \quad (1)$$

where RR_0 is the interval ending at beat i , RR_{-1} and RR_{-2} are the two intervals before it, and RR_{+1} and RR_{+2} are the two intervals after it. Intervals are in seconds, clipped to $[0.2, 3.0]$ s, and set to 0.8 s where they would extend past the start or end of a record. They are computed between consecutive entries of the database annotation file. Those entries include non-beat annotations such as rhythm-change, noise and artefact markers, so a small fraction of intervals are annotation-to-beat gaps rather than beat-to-beat intervals. §4.6 quantifies this artefact and its effect on every reported model. It affects all models identically, because every model receives the same features.

Temporal scope. Three properties make the task *offline* beat classification rather than streaming classification. The interval vector contains two intervals that follow beat i . The window’s 750 ms post-R segment contains the next annotated beat for 48.8% of DS2 beats. The filter is zero-phase and applied per record. We therefore do not describe the model as real-time or streaming. A deployment would need a lookahead buffer and a causal filter.

3.2. Inter-Patient Split

We use the DS1/DS2 partition of de Chazal *et al.* [7], splitting DS1 into training and validation records following Takalo-Mattila *et al.* [1]. DS1 training uses the 18 records 101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 215 and 220. DS1 validation uses the 4 records 208, 209, 223 and 230, which carry every model and architecture selection decision. DS2, held out, comprises the 22 records 100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233 and 234. The partition is by record, and record disjointness is asserted programmatically and covered by a regression test. One subject contributes to two partitions: records 201 (training) and 202 (DS2) were taken from the same analogue tape, a known property of the de Chazal partition that we retain unmodified.

Annotations are mapped to the five AAMI EC57 classes [5, 26] (Table 1). The Q class has only 15 beats in these records (6 in training, 7 in DS2), so no per-class conclusion can be drawn for it. Record 208 holds 372 of DS1’s 414 fusion beats and lies in the validation partition, which leaves 28 F beats for training. Fig. 1 summarises how each partition and each external database is used.

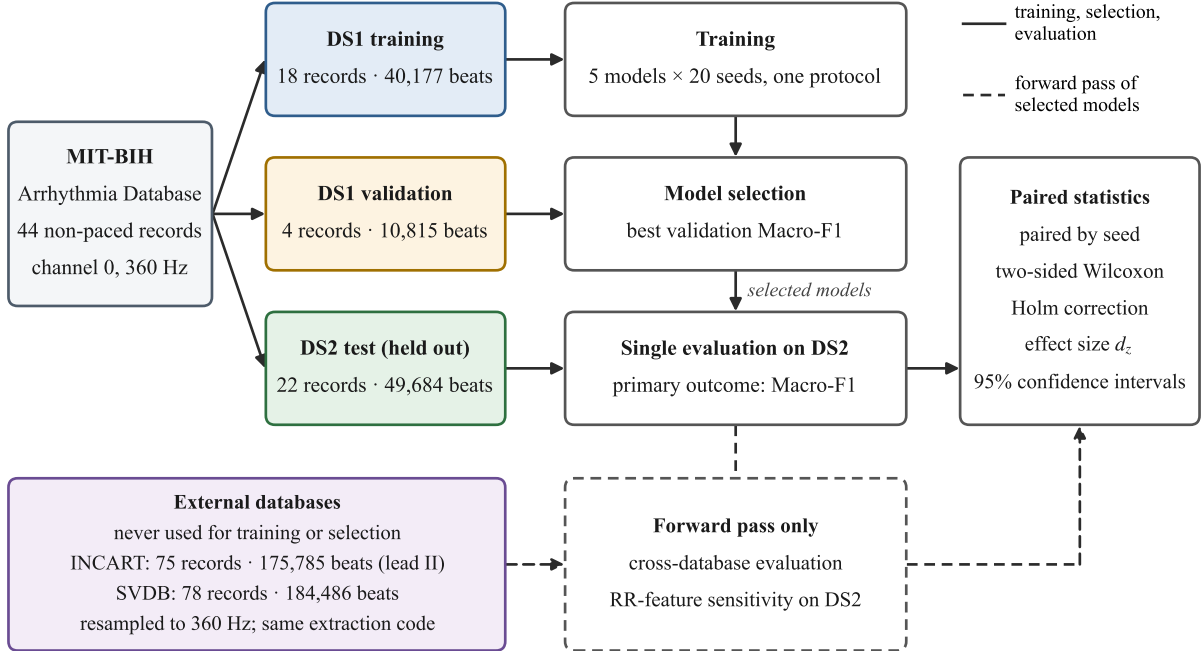


Figure 1: Evaluation protocol. Training records fit the models, the four validation records determine every checkpoint and configuration choice, and DS2 is evaluated once with the selected models. The external databases and the RR-feature sensitivity analysis use forward passes of the selected models only. Counts are read from the split definition and the extraction outputs.

Table 1: Class distribution and AAMI mapping.

Class	Included Annotations	Train	Val	Test (DS2)	Total	Ratio (vs N)
N	Normal, L, R, e, j	37,337	8,502	44,232	90,071	1.0 : 1
S	A, a, J, S	485	458	1,837	2,780	32.4 : 1
V	V, E	2,321	1,467	3,220	7,008	12.8 : 1
F	F	28	386	388	802	112.3 : 1
Q	/, f, Q	6	2	7	15	6004 : 1
Total		40,177	10,815	49,684	100,676	

3.3. PC-WavKAN

The model (Fig. 2) has a morphology branch and a rhythm branch whose outputs are concatenated and classified.

Morphology branch

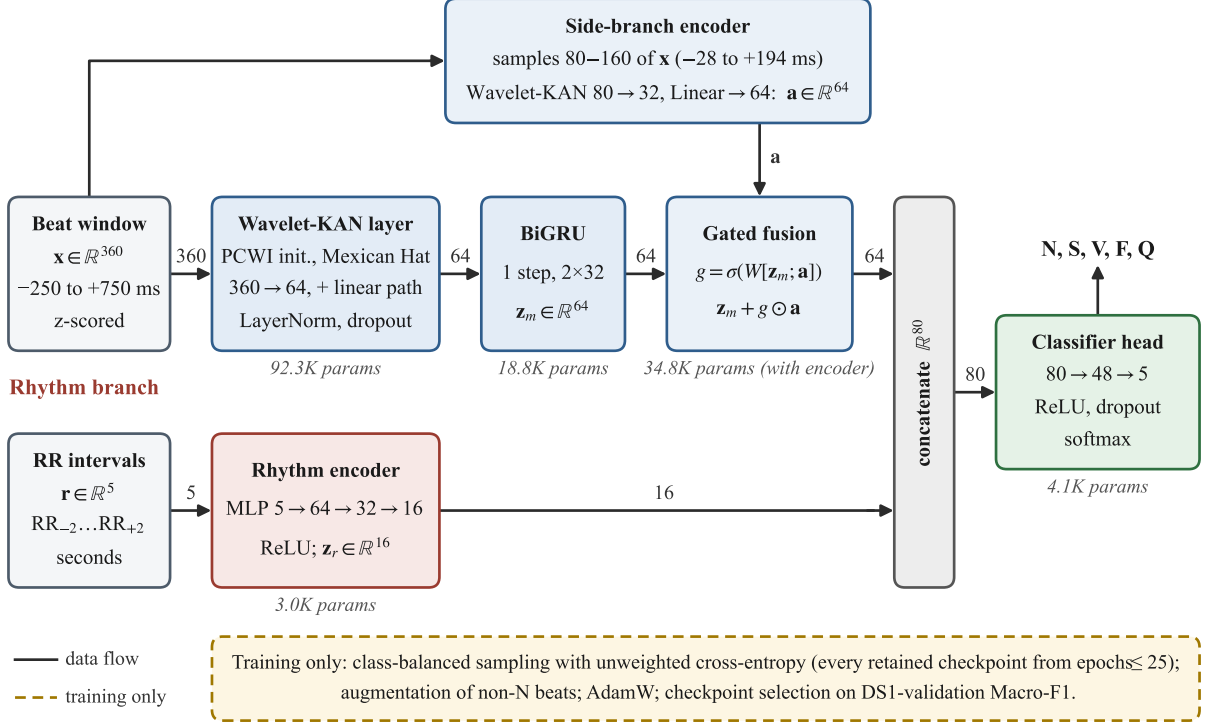


Figure 2: PC-WavKAN as implemented, with tensor dimensions and per-block trainable parameters (153,045 in total). Morphology branch: the beat window passes through a wavelet-KAN layer with prior-centred initialisation (PCWI) and a linear residual path, followed by layer normalisation, dropout and a bidirectional GRU applied to a single time step. A side branch encodes raw-beat samples 80–160 (28 ms before to 194 ms after the R-peak) into $\mathbf{a}$, which is added to the GRU output through a learned sigmoid gate g . Rhythm branch: an MLP encodes the five RR intervals. The two branch outputs are concatenated and classified by an $80 \rightarrow 48 \rightarrow 5$ head. The mechanisms in the dashed band act during training only.

Wavelet-KAN layer. Following Wav-KAN [9], every edge (j, k) carries its own Mexican Hat function $\psi(u) = (1 - u^2)e^{-u^2/2}$. The layer maps $x_i \in \mathbb{R}^{360}$ to $\mathbb{R}^{64}$:

$$z^{(k)} = \sum_{j=1}^{360} w_{j,k} \psi\left(\frac{x_j - \mu_{j,k}}{|\gamma_{j,k}|}\right) + \lambda \sum_{j=1}^{360} v_{j,k} x_j, \quad k = 1, \dots, 64, \quad (2)$$

with learnable translation $\mu_{j,k}$, dilation $\gamma_{j,k}$ and weight $w_{j,k}$, and a linear residual path with fixed gate $\lambda = 0.1$. ψ is evaluated on the *amplitude* x_j of input sample j , so μ and γ are an amplitude offset and an amplitude width. They are not a temporal position or duration. The output is layer-normalised, passed through dropout ($p = 0.2$), and fed to a bidirectional GRU with 32 units per direction. The 64 activations form a single token, so the GRU executes one step in each direction from a zero state. It therefore acts as a gated transformation of the channel vector and integrates no temporal context. It has 18,816 parameters (12.3% of the model).

Prior-centred wavelet initialisation (PCWI). The 64 output channels are split into three blocks, and every edge feeding block c is initialised around a fixed pair $(\bar{\mu}_c, \bar{\gamma}_c)$:

$$\mu_{j,k} \sim \mathcal{U}(\bar{\mu}_c - \epsilon_c, \bar{\mu}_c + \epsilon_c), \quad \gamma_{j,k} \sim \mathcal{U}(\bar{\gamma}_c - \delta_c, \bar{\gamma}_c + \delta_c). \quad (3)$$

Channels 1–32 use $(0.0, 0.05)$ with $(\epsilon, \delta) = (0.02, 0.005)$, channels 33–48 use $(-0.10, 0.12)$, and channels 49–64 use $(+0.10, 0.10)$, the last two with $(\epsilon, \delta) = (0.03, 0.01)$. The values were chosen with reference to the relative amplitude and sharpness of the QRS complex and the P and T waves on the z -scored scale, and we refer to the blocks by those names (QRS-, P- and T-prior blocks). Without PCWI, all edges use the isotropic initialisation $\mu \sim \mathcal{U}(-0.3, 0.3)$, $\gamma \sim \mathcal{U}(0.05, 0.20)$. The weights $w_{j,k}$ and $v_{j,k}$ use Kaiming-uniform initialisation in both cases.

Two properties limit what PCWI can encode. First, each block’s prior is shared by all 360 input samples, so no prior is tied to the temporal location of a waveform component. Second, the 64 channels are exchangeable. They feed a layer normalisation initialised identically across channels and a recurrent layer whose input weights are drawn independently and identically across channels. Permuting the channels together with the corresponding downstream weights leaves the network function unchanged, which we verified numerically (maximum output difference 10^{-7}). A network in which the P- and T-block priors are exchanged is therefore an exact relabelling of one in which they are not, with the same initial distribution. What PCWI determines is the set of initial (μ, γ) values, 32 channels narrow and centred at zero and 16 each slightly offset and wider, not an assignment of channels to waveform components. We name it *prior-centred* for that reason.

Gated side branch. A side branch reads raw-beat samples 80–160, which with the R-peak at sample 90 spans 28 ms before to 194 ms after the R-peak, i.e. the QRS complex and early ST segment. The branch was designed to cover the P-wave, but its sample range was fixed for an R-peak at sample 180, and it does not reach the P-wave in the window as extracted. We therefore describe it by what it reads. The samples pass through a randomly initialised wavelet-KAN layer ($80 \rightarrow 32$), are projected to 64 dimensions, and pass through a four-head attention block over a single key/value token. With one token the attention softmax is degenerate, so at inference the block is a learned linear map. A sigmoid gate then combines the result with the GRU output. The branch has 34,816 parameters.

Rhythm branch and classifier. The interval vector is encoded to 16 dimensions by an MLP ($5 \rightarrow 64 \rightarrow 32 \rightarrow 16$, ReLU). An alternative encoder that projects each interval to a 16-dimensional token and applies two-head self-attention was also evaluated (§4.1). The 80-dimensional concatenation is classified by an $80 \rightarrow 48 \rightarrow 5$ head. With the MLP rhythm encoder the model has 153,045 trainable parameters, and 154,325 with the self-attention encoder. All results outside §4.1 use the 153,045-parameter configuration.

3.4. Training

Table 2 collects the settings. Training uses AdamW (learning rate 10^{-3} , weight decay 10^{-4}), batch size 64, gradient-norm clipping at 1.0 and a cosine learning-rate schedule over a nominal 100-epoch budget. Early stopping monitors DS1-validation Macro-F1 with patience 15, and the checkpoint of the best validation epoch is retained. In the PC-WavKAN runs trained with class-balanced sampling, training stopped between epochs 17 and 40, so the learning rate changed little before the retained epoch. The baseline scripts did not log their stopping epoch.

Table 2: Training and evaluation settings. Every setting above the rule is shared by all five models; the settings below it differ by design (§3.5).

Setting	Value
Inputs	beat window (360 samples) and five RR intervals
Optimiser	AdamW; learning rate 10^{-3} ; weight decay 10^{-4}
Learning-rate schedule	cosine annealing over a nominal 100 epochs
Batch size; gradient clipping	64; gradient norm 1.0
Early stopping	patience of 15 epochs on DS1-validation Macro-F1
Retained checkpoint	epoch with the best DS1-validation Macro-F1
Seeds	20, identical for every model (§3.7)
Imbalance handling, PC-WavKAN	class-balanced sampling with unweighted cross-entropy
Imbalance handling, baselines	natural sampling; inverse-frequency weighted cross-entropy (S weight $\times 8$); focal loss without weights for CNN+Focal
Augmentation	PC-WavKAN only: non-N beats (noise, baseline wander, scaling)

Class-balanced sampling. For the first 25 epochs, mini-batches are drawn by a sampler that selects each class with equal probability, and the loss is unweighted cross-entropy. The training script then enters a second phase in which the sampler is annealed towards the natural class distribution and the loss becomes a class-weighted focal loss. In all 60 runs that used this schedule (three configurations, 20 seeds each), the retained checkpoint came from the first phase (best epochs 2–25), because validation Macro-F1 never improved after the switch. The second phase therefore did not contribute to any reported model. We describe and evaluate the procedure by what determined those models: class-balanced sampling with unweighted cross-entropy and validation-based checkpoint selection, abbreviated CBS. The comparator, used by all four baselines and by the no-CBS arms, is natural-distribution sampling with inverse-frequency class-weighted cross-entropy, with the S-class weight multiplied by 8.

Augmentation. During training only, every non-N beat in a mini-batch is perturbed with probability one. The perturbation is additive Gaussian noise at 25 dB SNR, a sinusoidal baseline wander of 0.5–2.5 Hz with amplitude 0.01–0.05 (in z -units), and amplitude scaling by $\mathcal{U}(0.9, 1.1)$, applied after z -scoring. The baselines are trained without augmentation (§3.5).

3.5. Baselines and What Is Matched

Four architectures spanning 29K–118K parameters are compared (Table 3). *Held identical:* data, preprocessing, split, input representation (every model receives both the beat window and the interval vector), optimiser, learning rate, weight decay, schedule, epoch budget, early-stopping rule and patience, batch size, gradient clipping, checkpoint-selection rule, the 20 seeds, and the class-weight formula. *Not held identical:* the imbalance mechanism and augmentation. The baselines use natural-distribution sampling with class-weighted cross-entropy, except CNN+Focal, which uses focal loss without static class weights. We also tested a class-balanced sampler combined with inverse-frequency weights for the baselines. At this imbalance ratio that combination drove the majority-class gradient to near zero, and three of the four baselines then never predicted class N, so it was not used. The comparison in §4.2 is therefore between complete methods, i.e. architecture together with its imbalance strategy, under a common protocol and budget. §4.4 and §4.5 quantify the two mechanisms on PC-WavKAN itself.

The baseline labelled B-Spline KAN is the same wavelet-KAN layer with a single cubic B-spline in place of the Mexican Hat on every edge, isotropic initialisation, no side branch, and the same GRU, rhythm MLP and head. It is not the grid-based spline KAN of Liu *et al.* [8].

Table 3: Baselines. Protocol, split, inputs, optimiser, budget, checkpoint rule and seeds are identical to PC-WavKAN (§3.5).

Model	Params	Imbalance handling
ResNet1D (4-block residual CNN, 32 channels)	62,869	Weighted CE
Transformer ($d_{\text{model}}=32$, 2 layers, 4 heads)	29,653	Weighted CE
CNN+Focal (4-layer 1D CNN)	71,061	Focal loss ($\gamma=2$)
B-Spline KAN (single B-spline per edge; no PCWI or side branch)	118,229	Weighted CE
PC-WavKAN (§3.3)	153,045	CBS + augmentation

3.6. Parameter-Space Prior Retention

To test RQ3 we measure how close the trained wavelet parameters remain to the PCWI prior values. For block c ,

$$\text{PPR}_c = \max\left(0, 1 - \frac{1}{2} \left[\frac{|\overline{\mu_{j,k}} - \bar{\mu}_c|}{\Delta_\mu} + \frac{||\overline{\gamma_{j,k}}| - \bar{\gamma}_c|}{\Delta_\gamma} \right] \right), \quad (4)$$

where the overlines denote means over all edges of block c , and $\Delta_\mu = 0.40$, $\Delta_\gamma = 0.20$ are fixed normalising spans. The reported PPR is the unweighted mean over the three blocks. The measure is computed from parameters alone and never observes a model output. Because PCWI initialises at the prior values, an untrained PCWI model scores near-maximally by construction. PPR is therefore interpretable only against reference points, and we report three: untrained with PCWI, untrained with isotropic initialisation, and *trained* with isotropic initialisation. The last one tests whether training moves parameters towards the prior values. Per-edge absolute deviations are averaged rather than block means, because per-edge drift is close to symmetric and largely cancels in the mean (in the trained model, mean per-edge $|\Delta\mu| = 0.064$ against 0.004 for the deviation of the block means).

3.7. Statistical Methodology

Every arm is trained with the same 20 seeds: 42, 101, 777, 2026, 9999, 1234, 2024, 31415, 27182, 7, 11, 13, 888, 555, 333, 99, 1001, 5050, 8080 and 1998. Comparisons are paired by seed label. Within one architecture, the same seed gives the same initialisation. Across architectures the pairing is by label only and implies no shared initialisation or data order. The signed-rank test remains valid in that case under the null hypothesis of identical distributions. Significance is assessed by the two-sided Wilcoxon signed-rank test [27], with Holm–Bonferroni correction [28] within declared families. The metrics of one arm-versus-arm comparison form one family, as do the comparisons against the four baselines on one metric. All p -values are Holm-adjusted unless labelled raw. Effect sizes are the paired standardised mean difference $d_z = \bar{\delta}/s_\delta$, reported against Cohen’s conventional thresholds [29]. **Positive d_z favours PC-WavKAN, the adopted configuration, or the treatment arm.** We also report 95% confidence intervals on the paired mean difference. At $n = 20$ and $\alpha = 0.05$ two-sided, a paired test has 80% power only for $d_z \gtrsim 0.66$, and non-significant results are reported with their intervals rather than as

evidence of absence. No equivalence margin was specified, so parity results are not equivalence claims.

Outcome hierarchy. Results are reported in §4. The *primary* outcome is DS2 five-class Macro-F1 of the adopted configuration against the four baselines. The *pre-specified secondary* outcome is the effect of the imbalance mechanism on per-class recall, tested on the base configuration (§4.4). Everything else is *exploratory*. That includes the same comparison on the adopted configuration, which uses arms trained for the augmentation analysis, and the interval ablation, the cross-database evaluation, the prior-retention analysis and the artefact sensitivity analysis. The last four, and the four-class Macro-F1, were added during a final audit after the primary results were known, and are labelled as such where reported.

Software. Models are implemented in PyTorch. Every quantitative claim in this paper is re-derived from the per-seed result files by a verification script released with the code (Code availability).

4. Results

4.1. Architecture Selection on Validation Data

The configuration reported in the rest of this paper was selected using DS1 validation only. Table 4 gives the single-variable ablation. The base configuration has the wavelet-KAN layer with PCWI, the gated side branch, the self-attention rhythm encoder and CBS, and each row changes exactly one component. The validation score of each seed is taken at the epoch its checkpoint was saved.

Table 4: Single-variable component ablation on **DS1 validation**, 20 seeds per row. Base configuration: PCWI, gated side branch, self-attention rhythm encoder and CBS; validation Macro-F1 0.4145 ± 0.0201 . Δ and d_z are variant minus base, so positive values favour the variant. p is Holm-adjusted across the seven comparisons.

Variant	Val. Macro-F1	Δ vs base	d_z	Holm p	Reading
No PCWI (isotropic init.)	0.3932 ± 0.0129	-0.0213	-1.05 (large)	0.0001	PCWI helps in this configuration
No gated side branch	0.3976 ± 0.0184	-0.0169	-0.54 (medium)	0.164	not resolved
No CBS (reweighting)	0.4053 ± 0.0173	-0.0092	-0.37 (small)	0.624	not resolved
Basis: Morlet	0.4219 ± 0.0187	$+0.0074$	$+0.31$ (small)	0.568	not resolved
Basis: DOG	0.4041 ± 0.0123	-0.0104	-0.45 (small)	0.164	not resolved
Basis: B-spline	0.4116 ± 0.0151	-0.0029	-0.14 (negl.)	0.624	not resolved
MLP rhythm encoder	0.4397 ± 0.0140	$+0.0252$	$+1.15$ (large)	0.0008	adopted

Three findings follow. *Initialisation:* replacing PCWI with the isotropic initialisation costs 0.021 validation Macro-F1 ($d_z = -1.05$, Holm $p = 1 \times 10^{-4}$), the only corrected positive effect among the components ablated. The comparison contrasts one structured initialisation with one isotropic distribution, and §3.3 shows that what PCWI fixes is a set of amplitude offsets and widths, not a physiological assignment. The finding is therefore that this initialisation helps, not that physiological content does. It was measured in the base configuration and was not re-measured on the adopted configuration. *Mother wavelet:* substituting Morlet, DOG or a B-spline for the Mexican Hat changes validation Macro-F1 by at most 0.010, and no difference approaches significance. *Rhythm encoder:* replacing self-attention with an MLP over the same five intervals improves validation Macro-F1 by 0.025 ($d_z = +1.15$, Holm $p = 8 \times 10^{-4}$). This configuration is the best of all eight on validation. It outperforms each of the seven alternatives (Holm-adjusted $p \leq 0.0032$ over that family), and we adopt it.

Test-set exposure. Table 4 alone determines the selection. DS2 metrics for non-adopted variants were nevertheless computed during development, and the DS2 ranking agrees with the validation ranking (adopted 0.357 ± 0.019 against 0.323 ± 0.018 for the base). DS2 was therefore not untouched over the course of the study, and §4.2 should be read as a held-out estimate under a validation-determined selection rule rather than a single-shot test. Its conclusion is a null result, which selection on DS2 would not be expected to produce. On arrays regenerated from PhysioNet with the released code, all 20 adopted-configuration checkpoints reproduce their DS2 predictions beat for beat, and 19 reproduce exactly the validation Macro-F1 logged at their saved epoch. Seed 1001’s logged validation score does not reproduce. Excluding it changes no corrected conclusion, though the unadjusted interval against B-Spline KAN (Table 5) then narrowly excludes zero ($[-0.031, -0.0002]$).

4.2. Primary Comparison

Table 5: **Primary outcome.** DS2 five-class Macro-F1, PC-WavKAN versus four baselines, 20 paired seeds. Δ , CI and d_z are PC-WavKAN minus baseline; p is Holm-adjusted across the four comparisons.

Model	Params	Macro-F1	Δ	95% CI of Δ	d_z / Holm p
ResNet1D	62,869	0.351 ± 0.019	+0.006	$[-0.008, +0.020]$	+0.19 / 1.00
Transformer	29,653	0.353 ± 0.040	+0.004	$[-0.013, +0.022]$	+0.12 / 1.00
CNN+Focal	71,061	0.362 ± 0.015	−0.005	$[-0.017, +0.006]$	−0.22 / 1.00
B-Spline KAN	118,229	0.371 ± 0.024	−0.014	$[-0.029, +0.001]$	−0.43 / 0.36
PC-WavKAN	153,045	0.357 ± 0.019	—	—	—

PC-WavKAN’s DS2 Macro-F1 is not statistically distinguishable from any baseline (Table 5). All Holm-adjusted $p \geq 0.36$, all effects are negligible to small, and every 95% interval on the paired difference contains zero, the widest bound being 0.029. The data exclude a large difference in either direction. They do not establish equivalence, for which no margin was set. PC-WavKAN is also the largest of the five models. Because the Q class contributes a near-constant zero (§4.3), we also report the four-class N/S/V/F Macro-F1 (exploratory): PC-WavKAN 0.446 ± 0.023 , ResNet1D 0.439 ± 0.023 , Transformer 0.441 ± 0.050 , CNN+Focal 0.453 ± 0.019 and B-Spline KAN 0.463 ± 0.030 , with all Holm-adjusted $p \geq 0.36$. The conclusion is unchanged.

Fig. 3 shows the structure behind these values, averaged over the 20 seeds. N and V separate well, each leaking under 6% into the other, with V-recall 0.898. S beats are split almost equally between V (0.40) and N (0.39), and most F beats are predicted as N (0.78).

4.3. Per-Class Operating Points

Table 6 gives precision, recall and F1 per class over 20 seeds, recomputed from stored per-seed predictions. The per-class F1 values average to the Macro-F1 of Table 5 for all five models.

The S-class operating point is poor in both directions. Recall is 0.198 at precision 0.175, so roughly four in five predicted S beats are wrong and four in five true S beats are missed. On S-F1, PC-WavKAN is better than ResNet1D and CNN+Focal, indistinguishable from the Transformer, and worse than B-Spline KAN ($d_z = -0.66$, Holm $p = 0.027$). The parity of §4.2 is therefore not uniform across classes. Q contributes an exact zero for every model, since 6 training beats cannot support a learned class. This caps the attainable five-class Macro-F1 at 0.8 and should be recalled whenever five-class inter-patient Macro-F1 values are compared.

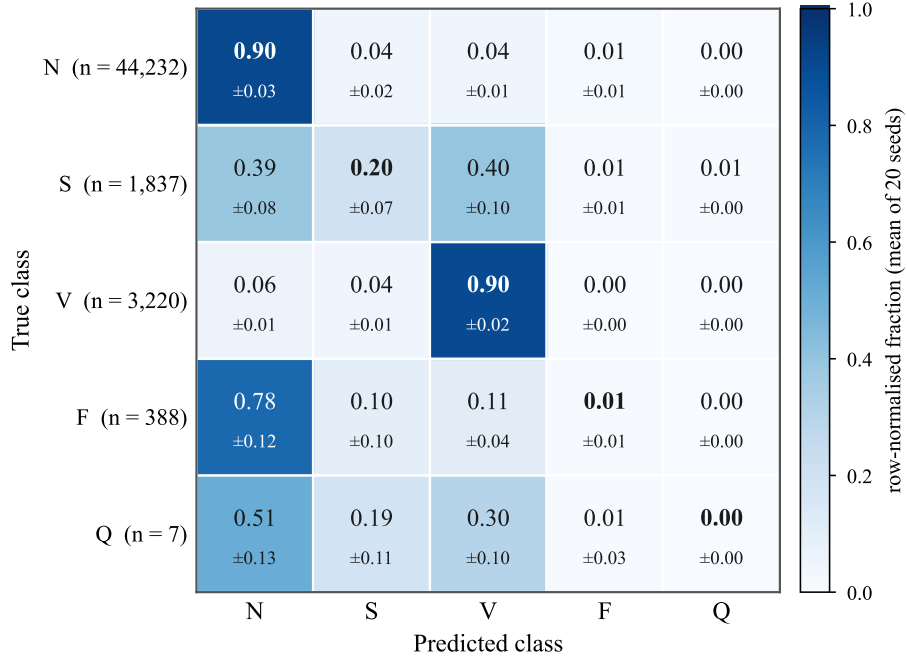


Figure 3: Row-normalised DS2 confusion matrix of PC-WavKAN, averaged over the 20 seeds. Each seed’s matrix is normalised by row and then averaged, and the across-seed standard deviation is shown under each value. Rows are labelled with their support; the Q row rests on 7 beats. The diagonal equals the per-class recall of Table 6.

Table 6: Per-class precision, recall and F1, 20 seeds, DS2. (a) PC-WavKAN, all classes. (b) S class across models, with the Holm-adjusted paired comparison of S-F1 against PC-WavKAN (positive d_z favours PC-WavKAN). S-class differences between models combine architecture and imbalance mechanism (§3.5). Q scores exactly zero for every model.

(a) PC-WavKAN, per class				
Class	Support	Precision	Recall	F1
N	44,232	0.971 ± 0.004	0.905 ± 0.025	0.936 ± 0.013
S	1,837	0.175 ± 0.079	0.198 ± 0.067	0.180 ± 0.064
V	3,220	0.528 ± 0.051	0.898 ± 0.020	0.663 ± 0.039
F	388	0.005 ± 0.006	0.006 ± 0.006	0.005 ± 0.006
Q	7	0.000	0.000	0.000
(b) S class across models				
Model	S-Prec.	S-Rec.	S-F1	d_z / Holm p
PC-WavKAN	0.175	0.198	0.180	—
ResNet1D	0.173	0.100	0.124	+0.77 / 0.004
Transformer	0.219	0.229	0.207	−0.35 / 0.312
CNN+Focal	0.306	0.075	0.115	+0.85 / 0.002
B-Spline KAN	0.237	0.280	0.244	−0.66 / 0.027

Comparison with the protocol’s source. de Chazal *et al.* [7], whose partition we adopt, report SVEB sensitivity 0.759 at positive predictivity 0.385, and VEB sensitivity 0.777 at positive predictivity 0.819. On the S class their linear-discriminant system is substantially better than every model trained here on both axes. On the V class our recall of 0.898 exceeds theirs, while our precision of 0.528 is lower. Their system uses two leads, hand-engineered morphology at fiducial points and interval features normalised to a record-level average, where ours uses one lead and a fixed window. We have not rerun their method. We report the comparison because the S-class gap is large and is against the protocol’s own source.

4.4. Class-Balanced Sampling Versus Reweighting

The pre-specified test compares CBS with reweighting on the base configuration, with augmentation in both arms. S-recall rises from 0.079 ± 0.016 to 0.123 ± 0.052 (Holm $p = 0.0016$ across five metrics, $d_z = +0.77$). On DS1 validation it rises from 0.218 ± 0.048 to 0.305 ± 0.047 ($d_z = +1.36$, raw $p = 2.4 \times 10^{-4}$). Table 7 repeats the comparison on the adopted configuration. It uses the two augmentation-free arms trained for §4.5, so the analysis is exploratory.

Table 7: CBS versus reweighting on the adopted configuration, both arms without augmentation, 20 paired seeds (exploratory). DS2 rows: d_z and p are CBS minus reweighting, Holm-adjusted across the five metrics. Validation rows: values at each run’s retained epoch, raw p .

Metric	Reweighting	CBS	d_z	p
<i>DS2 (held out)</i>				
Macro-F1	0.351 ± 0.014	0.364 ± 0.021	+0.52 (medium)	0.098
S-Recall	0.137 ± 0.037	0.222 ± 0.057	+1.47 (large)	2.9×10^{-5}
F-Recall	0.0003	0.0022	+0.88 (large)	0.013
V-Recall	0.880 ± 0.024	0.880 ± 0.017	−0.00 (negl.)	0.756
N-Recall	0.927 ± 0.021	0.917 ± 0.024	−0.32 (small)	0.228
<i>DS1 validation</i>				
Macro-F1	0.4285 ± 0.0125	0.4195 ± 0.0098	−0.57 (medium)	0.027 (raw)
S-Recall	0.2639 ± 0.0510	0.2391 ± 0.0425	−0.53 (medium)	0.025 (raw)

On DS2, CBS raises S-recall from 0.137 to 0.222, with a 95% confidence interval of $[+0.058, +0.112]$ on the paired difference. The absolute F-recall gain is negligible ($0.0003 \rightarrow 0.0022$). V- and N-recall show no corrected change. On the validation partition the direction reverses. CBS is nominally worse on both validation Macro-F1 and S-recall, although after Holm correction across the two validation metrics neither difference is significant ($p \approx 0.05$). The DS2 gain in S-recall is therefore robust across two configurations but is not reproduced on the four-record validation partition for the adopted configuration. We read this as evidence that the benefit of class-balanced sampling for supraventricular beats depends on the patients evaluated, and we do not claim it as a general property. The pre-specified test on the base configuration is the confirmatory evidence, and it is modest in absolute terms: an S-recall of 0.123.

4.5. Augmentation

PC-WavKAN is trained with minority-class augmentation and the baselines are not. On validation, augmentation is supported: with CBS in both arms, it raises validation Macro-F1 by 0.020 ($d_z = +1.16$, raw $p = 1.7 \times 10^{-4}$), consistent with its place in the adopted recipe. On DS2, removing it leaves Macro-F1 statistically unchanged (0.357 with

augmentation and 0.364 without, Holm $p = 0.46$) and reduces F-recall (0.0061 to 0.0022, Holm $p = 0.043$). The augmentation asymmetry therefore does not explain the parity in Table 5. Under an isolated 60-epoch recipe without CBS, the three noise-based strategies raised V-recall and lowered Macro-F1, and SMOTE produced the largest S-recall gain (0.316 $\rightarrow$ 0.648) at the largest Macro-F1 cost. That penalty did not carry over to the full recipe, which cautions against extrapolating short single-mechanism ablations.

4.6. Interval Features, Lookahead and an Annotation-Derived Artefact

An exploratory leave-one-out analysis masked each interval position at inference. It used the base configuration, whose rhythm features are identical to the adopted one’s, with 20 seeds and Holm correction across the five positions. Masking RR_0 caused by far the largest loss of S-recall (-0.0323 ± 0.0492 , Holm $p = 3 \times 10^{-4}$). This is expected, since a shortened preceding interval is the classical sign of a premature beat. Masking RR_{+2} caused a small loss that survives correction (-0.0045 ± 0.0103 , Holm $p = 0.014$). The other three positions had no corrected effect. The model therefore draws signal from an interval that has not elapsed when beat i occurs, which is one reason we treat the task as offline (§3.1).

Annotation-derived artefact. Because the intervals are computed between consecutive annotation-file entries, including non-beat markers, 4,578 of 49,684 DS2 beats (9.2%) have at least one interval that differs from the beat-to-beat value, and 1,433 (2.9%) have a different RR_0 . By class, RR_0 differs for 6.4% of V beats, against 2.7% of N beats and 1.3% of S beats, and the recorded value is almost always shorter than the true interval, which mimics prematurity. A rhythm-change marker is placed where an arrhythmic episode begins, so this artefact carries annotation-derived information that would not be available on unannotated recordings, and it correlates with the label. We quantified its effect without retraining. We recomputed every DS2 interval between beat annotations only and re-evaluated all 100 published checkpoints (five models, 20 seeds). The recomputation leaves the beats and their order unchanged. As a check, the same re-evaluation with the published intervals reproduced every model’s reported Macro-F1. Of the 100 checkpoints, 88 reproduced every DS2 prediction exactly. The remaining 12, all CNN+Focal, differed in 1 to 4 of 49,684 beats, which we attribute to floating-point differences of GPU inference. With beat-to-beat intervals, the Macro-F1 of every model rose slightly, in all 20 seeds. The mean increase was +0.002 to +0.005 across the five models (PC-WavKAN: 0.357 $\rightarrow$ 0.360), and PC-WavKAN’s V-recall changed by -0.001 . The primary comparison was unchanged: all Holm-adjusted $p \geq 0.38$, and every paired interval still includes zero. The artefact therefore did not inflate the reported performance and does not affect the comparative conclusions. The models were trained with it, and whether training on beat-to-beat intervals would change them was not tested.

4.7. Retention of the Prior Values

Table 8 evaluates PPR against its three reference points.

Training moves the PCWI parameters substantially: PPR falls from 0.973 to 0.783, and the mean per-edge translation deviation grows fivefold. The trained PCWI basis nonetheless remains closer to the prior values than an untrained isotropic basis (0.783 against 0.670). The trained isotropic row decides the interpretation. When training starts from the isotropic initialisation, it moves the parameters *further* from the prior values (0.670 $\rightarrow$ 0.579). Both comparisons separate the 20 models completely (two-sided Mann–Whitney $p = 7 \times 10^{-8}$). The prior values are therefore not where gradient descent

Table 8: Parameter-space Prior Retention (4), mean $\pm$ std over 20 models, with mean absolute per-edge deviations from the prior values. Untrained rows use 20 initialisation seeds. The trained isotropic row uses the 20 checkpoints of the no-PCWI ablation arm (base configuration).

Model state	PPR	$ \Delta\mu $	$ \Delta\gamma $
Untrained, PCWI	0.9729 ± 0.0001	0.013	0.004
Trained, PCWI (PC-WavKAN)	0.7826 ± 0.0351	0.064	0.055
Untrained, isotropic	0.6698 ± 0.0008	0.161	0.051
Trained, isotropic	0.5794 ± 0.0173	0.177	0.080

leads. The trained PCWI model’s proximity to them reflects where it started, not a learned preference. Together with the exchangeability of the channel blocks (§3.3), this means the wavelet parameters of this model do not support a physiological interpretation. They support the modest statement that the initial structure is partly retained. Fig. 4 shows the per-edge parameters behind these values. The retention lies mainly in the translations, which stay centred on the prior values. The dilations of the P- and T-prior blocks spread to a distribution broadly similar to that of the isotropic arm. Automated interpretability scores can rate untrained models as highly as trained ones [30], and our results show that parameter-space scores for KANs need the same reference points.

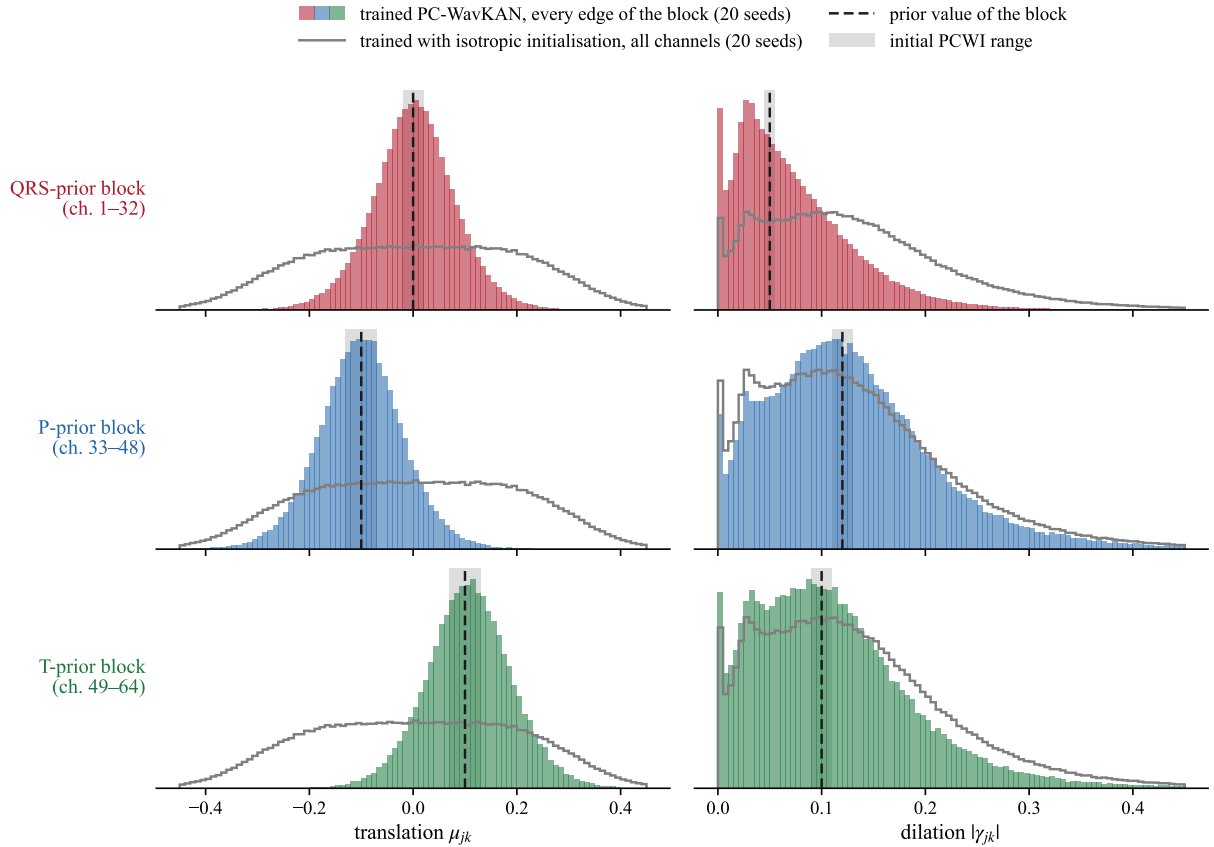


Figure 4: Per-edge wavelet parameters after training. Coloured histograms pool every edge of each channel block over all 20 seeds of PC-WavKAN. They are shown against the block’s prior value (dashed line) and initial PCWI range (grey band), and against the trained isotropic-initialisation arm, all channels (grey outline; the trained null of Table 8). Translations stay centred on the prior values but spread well beyond their initial range. The dilations of the P- and T-prior blocks spread to a distribution broadly similar to the null’s. Block names refer to initial values, not to functions.

4.8. Cross-Database Evaluation

Table 9 evaluates the five models trained on MIT-BIH on two further databases, forward pass only, 20 seeds each. The first is the St Petersburg INCART 12-lead Arrhythmia Database [25], all 75 records, lead II. The second is the MIT-BIH Supraventricular Arrhythmia Database (SVDB) [31], 78 records, channel 0, whose leads are not labelled. Both were resampled to 360 Hz and then passed through the same extraction code as the training data. A regression test confirms that this code reproduces the training arrays byte for byte. Neither database was used for any training, tuning or selection decision. SVDB was assembled to enrich supraventricular ectopy and serves as an S-class stress test. The analysis is exploratory.

Table 9: Zero-shot cross-database evaluation under the training preprocessing, 20 seeds per model. d_z and p compare PC-WavKAN with each baseline on Macro-F1 (positive favours PC-WavKAN), Holm-adjusted within each database.

Model	INCART (175,785 beats)				SVDB (184,486 beats)			
	Macro-F1	d_z / Holm p	V-Rec	S-Rec	Macro-F1	d_z / Holm p	V-Rec	S-Rec
PC-WavKAN	0.390 ± 0.017	—	0.832	0.671	0.275 ± 0.012	—	0.788	0.126
ResNet1D	0.431 ± 0.017	$-1.80 / < 0.001$	0.849	0.760	0.350 ± 0.026	$-2.49 / < 0.001$	0.832	0.256
Transformer	0.398 ± 0.032	$-0.24 / 0.245$	0.852	0.817	0.316 ± 0.031	$-1.36 / < 0.001$	0.820	0.305
CNN+Focal	0.431 ± 0.020	$-1.53 / < 0.001$	0.807	0.560	0.354 ± 0.013	$-4.86 / < 0.001$	0.769	0.179
B-Spline KAN	0.367 ± 0.017	$+0.83 / 0.003$	0.820	0.606	0.267 ± 0.016	$+0.33 / 0.231$	0.791	0.125

PC-WavKAN shows no transfer advantage. On INCART its Macro-F1 of 0.390 is significantly below ResNet1D and CNN+Focal (both 0.431; $d_z \leq -1.53$, Holm $p < 0.001$). It is indistinguishable from the Transformer and above B-Spline KAN ($d_z = +0.83$, Holm $p = 0.003$). On SVDB it is significantly below all three non-KAN baselines (Holm $p < 0.001$) and indistinguishable from B-Spline KAN, and the two KAN-family models have the lowest Macro-F1 of the five. Fig. 5 places the in-distribution and cross-database comparisons on one scale.

On SVDB, both KAN-family models also have the highest S→V confusion rates (0.431 and 0.451, mean over seeds, against 0.231–0.395 for the other models) and assign the largest share of beats to V (18.7% and 20.7%, against 8.4–17.6%). We report this as a descriptive pattern; the study does not identify its cause.

Sensitivity to preprocessing. An earlier evaluation of the same checkpoints used windows from only 30 INCART records, on lead I, and applied no filter. It ranked PC-WavKAN first on INCART (Macro-F1 0.374 against 0.316–0.365 for the baselines). Matching the preprocessing to the training data reversed that ranking. Cross-database comparisons of this kind are therefore sensitive to preprocessing details that are rarely reported, and we report only the matched evaluation.

4.9. Efficiency

Table 10 reports inference on an x86 CPU, single-threaded, for the seed-42 checkpoint. Values are the mean of three repeated runs of 1000 timed trials after a 20-iteration warm-up, and $\pm$ is the spread across runs.

Dynamic INT8 quantisation brings no benefit and slows inference. PyTorch’s dynamic quantisation converts `nn.Linear` modules only, while most of this model’s parameters sit in the custom wavelet-KAN layer and the GRU. The FP32 latency supports continuous offline processing on a general-purpose CPU. We made no ARM or on-device measurement and make no claim about wearable deployment.

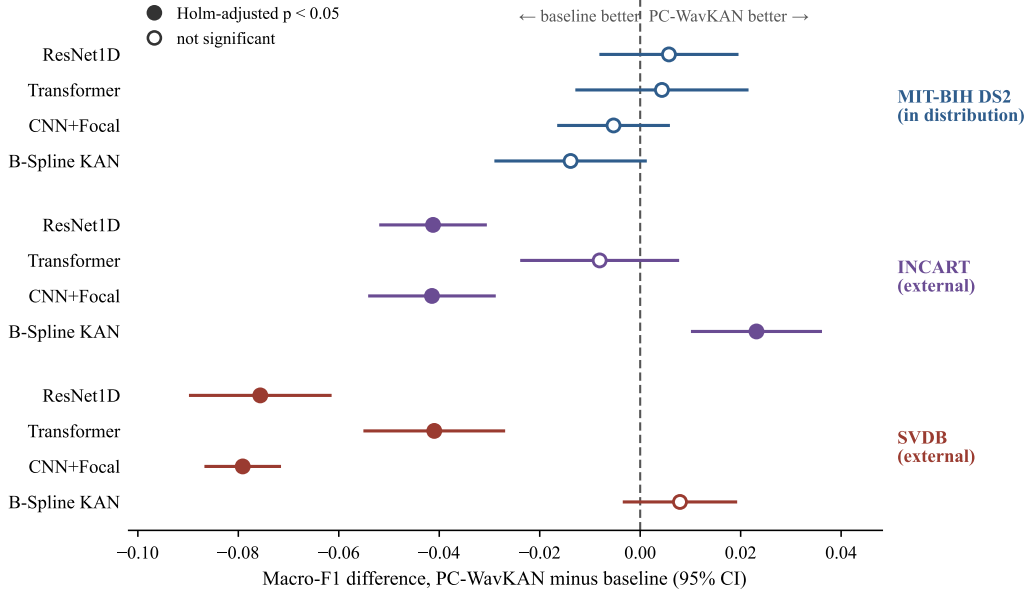


Figure 5: Paired Macro-F1 differences, PC-WavKAN minus each baseline, with 95% confidence intervals over 20 seeds. The panels cover MIT-BIH DS2 (Table 5) and the two external databases (Table 9). Filled markers indicate Holm-adjusted $p < 0.05$ across the four baselines within a database. In distribution, every interval includes zero. Out of distribution, PC-WavKAN is below both convolutional baselines.

Table 10: Measured inference, PC-WavKAN, x86 CPU, single thread, PyTorch 2.6.

Metric	Value
Model size, FP32 / INT8 dynamic	0.596 / 0.551 MB (1.08 $\times$)
Latency, batch 1, FP32	2.93 ± 0.38 ms/beat (median 2.78)
Latency, batch 1, INT8 dynamic	4.00 ± 0.35 ms/beat (0.73 $\times$, slower)
Latency, batch 64, FP32	0.429 ms/beat
Peak resident memory	235 MB

5. Discussion

What the study establishes. Under a matched protocol with multiplicity control, the wavelet KAN is not statistically distinguishable from four standard architectures on Macro-F1, with every paired interval within ± 0.03 . It is not more accurate than a 30K-parameter Transformer, and we do not present the wavelet-KAN basis as an accuracy mechanism. Among its design choices, initialising the wavelet parameters in a structured, prior-centred way helped relative to an isotropic initialisation, and the choice of mother wavelet did not measurably matter. The inspectability argument for wavelet KANs does not transfer to a physiological reading of this model’s parameters. The channel blocks are exchangeable, and training moves unconstrained parameters away from the prior values, not towards them. An inspectable parameter is not thereby an interpretable one. A claim that such parameters encode physiology would need behavioural evidence, for example a selective loss of one class when one block’s functions are perturbed. The null models we report are the minimum needed to interpret a parameter-space score.

Why self-attention over interval features fails. Removing self-attention from the rhythm encoder improved validation Macro-F1 with a large effect, while §4.6 shows the interval features carry real signal. The problem is therefore the encoder, not the features. Self-attention is close to permutation-equivariant without positional information, whereas the information in the interval vector is concentrated in one *position*, RR_0 . An MLP over a fixed-length vector has that positional distinction for free. We observed this in one architecture and one dataset. It is a caution worth checking for short fixed-length feature vectors, not a general result.

The external evaluation qualifies the parity. Under preprocessing matched to training, the wavelet KAN transferred worse than both convolutional baselines on both external databases, and the two KAN-family models were the weakest on SVDB. In this study, therefore, the parameterisation that is inspectable in principle came with no accuracy advantage in distribution and a disadvantage out of distribution. Anyone choosing an architecture for data from other sources should weigh the second against the first. The earlier, unmatched evaluation that appeared to show the opposite on INCART is a concrete example of how a preprocessing mismatch can manufacture an apparent transfer advantage.

5.1. Limitations

- (1) Parity, not advantage or equivalence.** No equivalence margin was set. The intervals bound a true Macro-F1 difference at roughly ± 0.03 .
- (2) Absolute minority-class performance.** S-recall is 0.198 at precision 0.175. F-recall is negligible in every configuration (0.0003 to 0.0061), reflecting 28 training beats. Q, with 6 training beats, supports no per-class conclusion and adds a near-constant zero to the five-class average.
- (3) Selection history of the test set.** The adopted configuration is determined by validation data, but DS2 metrics of non-adopted variants were computed during development.
- (4) Scope of the component evidence.** Every component ablation, including PCWI, was run on the base configuration. The adopted configuration differs from it only in the rhythm encoder, and no other component was re-ablated on it. We do not claim that PCWI improves the adopted model.

(5) Validation partition. Validation comprises four records, and one record holds 90% of DS1’s fusion beats. Selection decisions and the validation reversal in Table 7 should be read with that in mind.

(6) Non-causal inputs and an annotation-derived artefact. The inputs include two following intervals, a window that often contains the next beat, and a zero-phase record-level filter. The interval features also carry an annotation-derived artefact in 9.2% of DS2 beats. Its effect is bounded in §4.6. It was not removed by retraining.

(7) External data. Cross-database evidence rests on two databases. INCART is evaluated on lead II rather than MIT-BIH’s modified lead II, and SVDB’s leads are unlabelled. PTB-XL [32] was processed but is excluded. Its labels are record-level, not beat-level. Our extraction used a symmetric window with the R-peak at sample 180 rather than 90, no filter, R-peaks from a peak detector rather than expert annotations, and interval features expressed as ratios to a local 10-beat mean rather than in seconds. A Macro-F1 of 0.200 ± 0.005 was obtained, and we attribute it to these mismatches rather than to transfer.

(8) Baselines and hardware. The B-Spline KAN baseline is a single-bump variant, not a grid-based spline KAN. Latency is x86 CPU only.

Future work. The results point to three specific questions: whether a causal variant (preceding intervals only, a truncated window and a causal filter) preserves the parity; whether the wavelet-KAN family’s behaviour holds on 12-lead recordings with beat-level labels; and whether any behavioural test links individual wavelet parameters to class-specific evidence.

6. Conclusion

We evaluated a 153,045-parameter wavelet Kolmogorov–Arnold network for inter-patient ECG heartbeat classification under the de Chazal protocol, with 20 paired seeds, multiplicity correction, effect sizes, confidence intervals and architecture selection on validation data. Its DS2 Macro-F1 of 0.357 ± 0.019 is not statistically distinguishable from four standard baselines of 29K–118K parameters, and all five models share a poor supraventricular operating point. Structured initialisation of the wavelet parameters helped relative to an isotropic one, the mother wavelet did not measurably matter, and self-attention over interval features was harmful. The parameters do not support a physiological reading once exchangeability and a trained null are considered. Class-balanced sampling raised DS2 supraventricular recall but not on the validation partition. Under preprocessing matched to training, the wavelet KAN transferred worse than both convolutional baselines to INCART and SVDB. We report these null and negative findings alongside the positive ones because, on this evidence, they are the more transferable part of what the study established.

CRedit authorship contribution statement

Venkateramanan Manivannan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.
Ramanathan Lakshmanan: Conceptualization, Resources, Validation, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics statement

This study uses only publicly available, de-identified PhysioNet databases [25]. No new human-subject data were collected, and no institutional review board approval was required.

Data availability

All datasets are publicly available from PhysioNet [25]: the MIT-BIH Arrhythmia Database [24] (<https://physionet.org/content/mitdb/1.0.0/>), the St Petersburg INCART 12-lead Arrhythmia Database (<https://physionet.org/content/incartdb/1.0.0/>), the MIT-BIH Supraventricular Arrhythmia Database [31] (<https://physionet.org/content/svdb/1.0.0/>) and PTB-XL [32] (<https://physionet.org/content/ptb-xl/1.0.3/>).

Code availability

<https://github.com/vrhrs/WavKAN-CL> contains the source code, preprocessing and training scripts, statistical and figure code, the per-seed records of every arm reported here (test metrics, training histories, confusion matrices), the 20 checkpoints of the evaluated PC-WavKAN configuration, and the 20 checkpoints of its isotropic-initialisation ablation arm used for the prior-retention null. Baseline checkpoints are available from the authors on request. `src/verify_manuscript_numbers.py` re-derives the quantitative claims of this manuscript from those files. The repository name is historical.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with writing and code development. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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